

LESSON 7-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Rational Numbers on a Number Line**

Maryland Standard 6.NS.C.6c • Learning target – I can plot a rational number and its opposite on a number line.

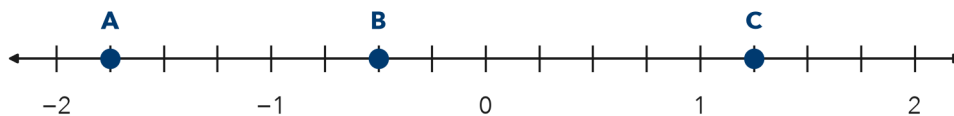
REMEMBER

- A **rational number** can be written as a fraction a/b of two integers ($b \neq 0$). Examples: $3/4$, -2.5 , 7 , $-1\frac{1}{2}$.
- Between two integers, split the gap into **equal parts** (halves, fourths, tenths).
- Negative numbers sit **left of 0**. -2.5 is halfway between -3 and -2 .

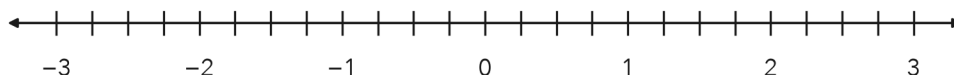
WORKED EXAMPLEPlot -1.75 and its opposite on a number line.**Step 1** -1.75 is between -2 and -1 . Split each unit into fourths; count 3 fourths left of -1 .**Step 2** The opposite, 1.75 , is 3 fourths right of 1 .**Answer points at -1.75 and 1.75** **WORD BANK****rational number** – a number you can write as a fraction**integer** – ..., -2 , -1 , 0 , 1 , 2 , ...**negative** – less than 0 ; to the left of 0 **between** – in the middle of two numbers**PRACTICE**

show your work • label units

1. Write the number for each point as a decimal or a fraction.

**a.** A = _____**b.** B = _____**c.** C = _____

2. Plot and label each number on the number line: -2.5 , $1/2$, -0.75 , $1\frac{1}{4}$

Step 1 Find the two integers on each side of the number.**Step 2** Split the gap into equal parts.**Step 3** Count the parts and mark the point.

3. Name the two integers each number is between.

a. -4.6 is between _____**b.** $7/2$ is between _____**c.** -0.3 is between _____**d.** $-9/4$ is between _____

4. Write a rational number for each situation.

a. a loss of \$3.50 ____

b. a gain of $2\frac{1}{2}$ ft ____

c. 4.5 degrees below
0 ____

5. **Pond.** A frog is 1.5 feet **above** the pond. A fish is 2.25 feet **below** the top of the pond. The top of the pond is 0.

Frame The ____ is closer to 0 because ____.

a. Write a number for the frog and for the fish. ____

b. Which one is closer to the surface? ____

6. **Find the Error.** Tyrell plotted -1.5 between -1 and 0 . Explain his mistake.

Frame Tyrell's mistake is ____ . -1.5 is between ____ and ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Find a ruler or measuring cup at home. Show someone where $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{3}{4}$ are.

LESSON 7-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Opposites of Rational Numbers**

Maryland Standard 6.NS.C.6a • Learning target – I can find the opposite of a rational number and explain what taking the opposite twice does.

REMEMBER

- **Opposites** are the same distance from 0: 2.5 and -2.5 .
- The opposite of a negative number is **positive**.
- Taking the opposite **twice** gives back the number you started with.

WORKED EXAMPLE

Start at $3/5$. Find its opposite. Then find the opposite of that.

Step 1 Opposite of $3/5$ is $-3/5$.

Step 2 Opposite of $-3/5$ is $3/5$.

Answer $-3/5$, then $3/5$

WORD BANK

opposite – same distance from 0, other side
 $-(-3)$ – the opposite of -3 , which is 3

distance from 0 – how far a number is from 0

PRACTICE

show your work • label units

1. Write the opposite of each number, or find the value.

a. 4.8 _____

b. $-2/3$ _____

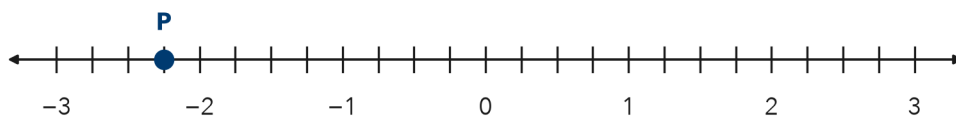
c. -0.05 _____

d. 0 _____

e. $-(-1.6) =$ _____

f. $-(-5/8) =$ _____

2. Point P is shown. Plot its opposite and label it Q.



Q = _____

3. **Game Score.** Lena has -12.5 points. A bonus card changes her score to the opposite number.

What is her new score? How far from 0 is it? _____

4. **Stretch.** Point M is 3.75 units from 0, to the left of 0. Point N is the opposite of M.

Step 1 M is to the left of 0, so M is negative.

Step 2 N is the opposite of M, so N is positive.

a. $M =$ _____

b. $N =$ _____

5. **Find the Error.** Kofi says $-(-2/5) = -2/5$. Explain his mistake. Give the correct value.

Frame Kofi is wrong because the opposite of a negative number is ____.

TEST PRACTICE

choose the best answer

6. What is the opposite of $-7/8$?

A $-7/8$

B $7/8$

C $8/7$

D $-8/7$

7. Point R is at 3.4. Where is the opposite of the opposite of R?

A -3.4

B 0

C 3.4

D 4.3

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #____
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AT HOME Explain to someone at home why $-(-1.5)$ is a positive number.